Chapter 8
The Bitcoin Network

Course Notes

CSE0469 Blockchain: Special Topics in SW Engineering

Yusuf Altunel

Based on Mastering Bitcoin 2e; Andreas M. Antonopoulos; O'Reilly, 2017

Content

- Peer-to-Peer Network Architecture
- Node Types and Roles
- The Extended Bitcoin Network
- Bitcoin Relay Networks
- Network Discovery
- Full Nodes
- Exchanging “Inventory”
- Simplified Payment Verification (SPV) Nodes

Content



Bloom Filters

How Bloom Filters Work

How SPV Nodes Use Bloom Filters



SPV Nodes and Privacy



Encrypted and Authenticated Connections

Tor Transport

Peer-to-Peer Authentication and Encryption



Transaction Pools

Peer-to-Peer Network Architecture



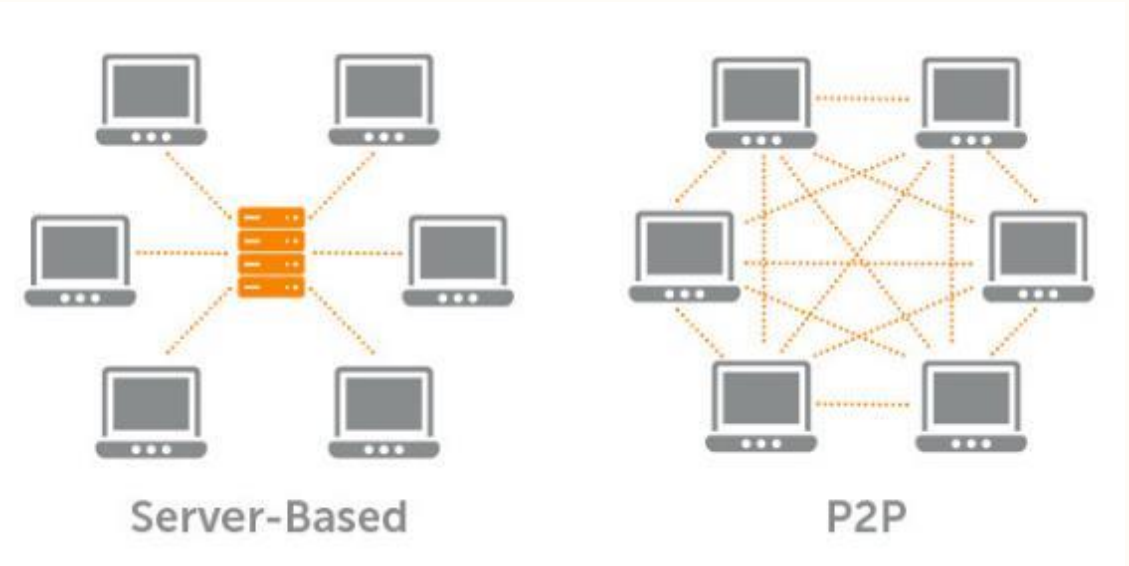
Mesh Network with Flat Topology



Topology Choice



Extended Protocol



Introduction

On top of Internet

Bitcoin is structured as a peer-to-peer network architecture

on top of the internet.

P2P

The term peer-to-peer, or P2P, means that

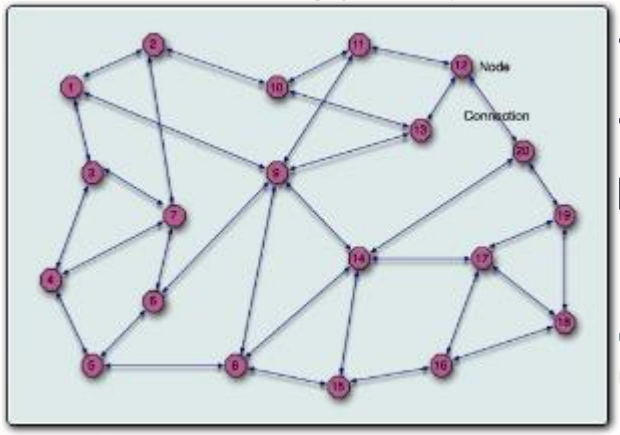
the computers that participate in the network

are peers to each other

Nodes are Equal

there are no “special” nodes,

all nodes share the burden of providing network services



Mesh Network with Flat Topology

Flat topology

The network nodes interconnect in a mesh network

with a “flat” topology.

Roles of Nodes

There is no server

no centralized service

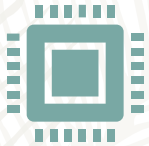
no hierarchy within the network.

Provide & consume services

Nodes in a P2P provide and consume services

acting as the incentive for participation.

Bitcoin's Topology Choice



Bitcoin is P2P

Bitcoin is a P2P digital cash system by design



the network architecture

is both a reflection

and a foundation of that core characteristic.



Core Design Principle

Decentralization of control is a core design principle



Achieved by flat & P2P

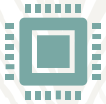
can only be achieved maintained by consensus network

flat

decentralized

P2P

Other Protocols on Bitcoin Network



Bitcoin network

the collection of nodes

running the **bitcoin P2P protocol**.



Mining: Stratum

there are other protocols such as Stratum



Stratum usage

for mining

lightweight or mobile wallets



Gateway

These additional protocols are provided

by gateway routing servers



routing servers

access the bitcoin network

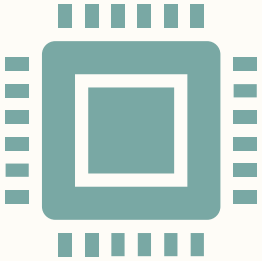
using the bitcoin P2P protocol



Extensions

extend that network to nodes running other protocols.

Extended Bitcoin Network



Stratum

Stratum servers connect Stratum mining nodes

to the main bitcoin network



via the Stratum protocol

bridge the Stratum protocol

to the bitcoin P2P protocol

Extended Bitcoin Network



“extended bitcoin network”

refer to the overall network

includes the bitcoin P2P protocol and extensions



Extensions

pool-mining protocols

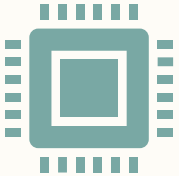
the Stratum protocol

any other related protocols



- Node Types
- Roles

Roles of Nodes



Node

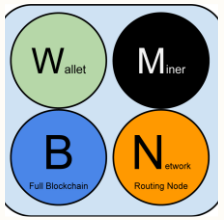
is Collection of Functions



Roles

depends on the functionality

Each node supporting



Possible Functions

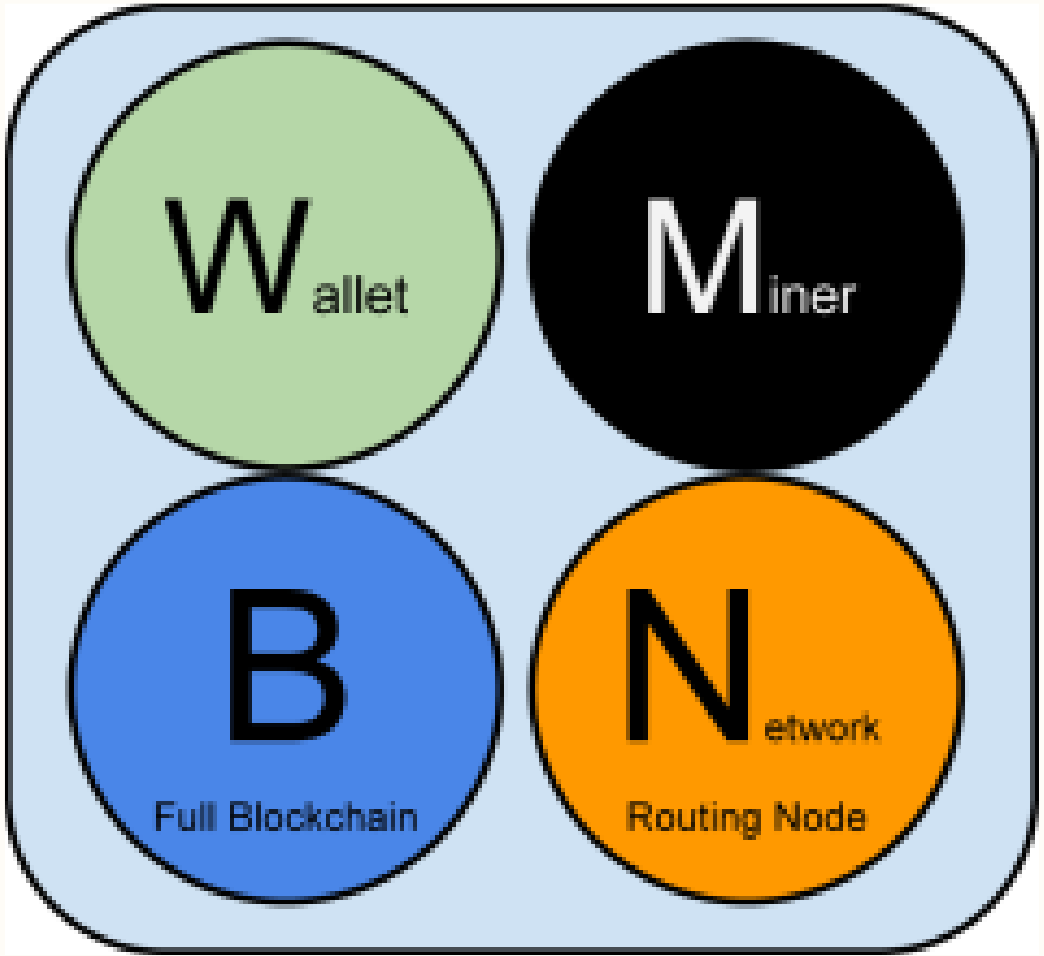
routing

keeping blockchain database

mining

wallet services

A bitcoin network node with four functions



Nodes Functionality



Routing

Nodes are routers

All nodes has routing function



Validation

transactions

blocks



Propagation

transactions

blocks



Discovery

maintain connections with peers

Full Nodes & Lightweight Nodes



full node

entire blockchain

up-to-date

complete full copy



Abilities

verify any transaction

without external reference

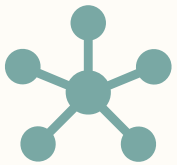


Authority

autonomous

authoritative

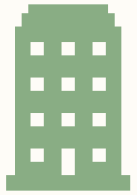
Full Nodes & Lightweight Nodes



Lightweight

only a subset

of the blockchain



SPV

simplified payment verification



Authority

verify transactions

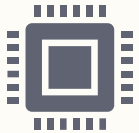
using **SPV** methods

Mining Nodes



Mining nodes

compete to create new blocks



Hardware

specialized hardware



ASICs

Application-specific Integrated Circuit



Functionality

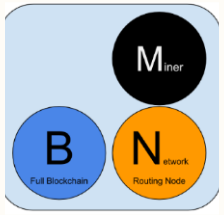
solve

the Proof-of-Work algorithm

Mining Nodes



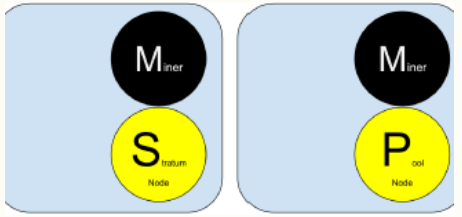
Mining nodes



Full Node Mining Nodes

maintain

full copy of the blockchain

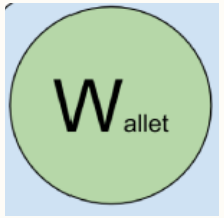


Lightweight Mining Nodes

participating in pool mining

depend on a pool

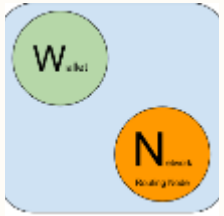
Wallet Nodes



Wallet Nodes

Wallet application

With or without Full Node



Desktop wallet

desktop bitcoin wallet application

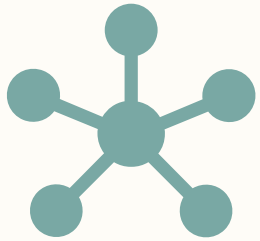


Smartphone Wallets

Wallet running on smartphones

SPV node

Other Nodes



Additional Nodes

servers

nodes running other protocols



Protocols

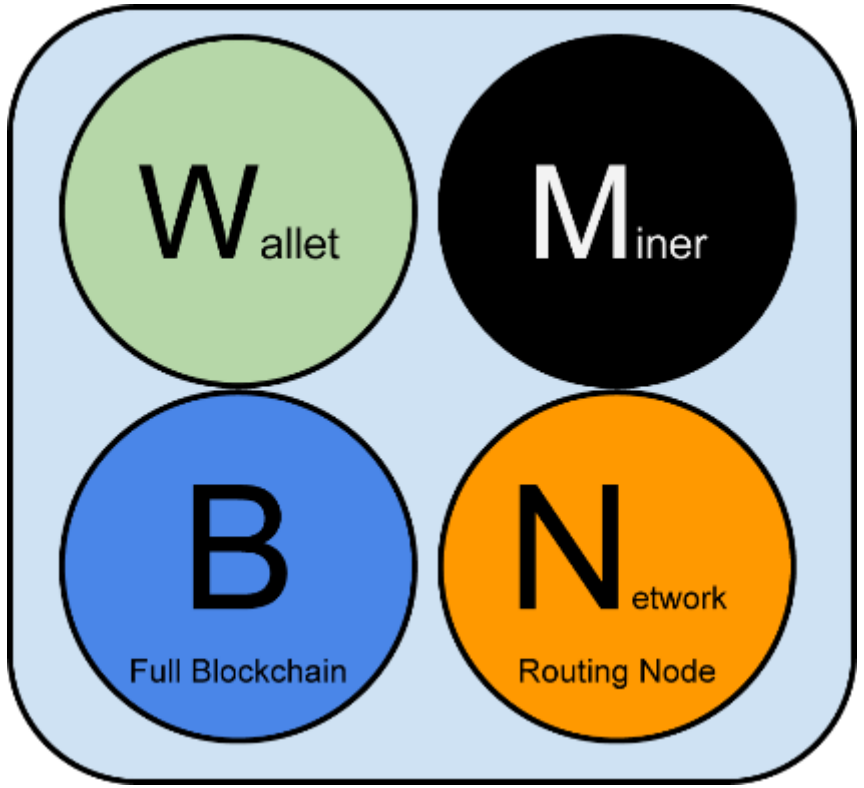
specialized mining pool protocols

lightweight client-access protocols



Different types of nodes on the extended bitcoin network

Figure 8-2

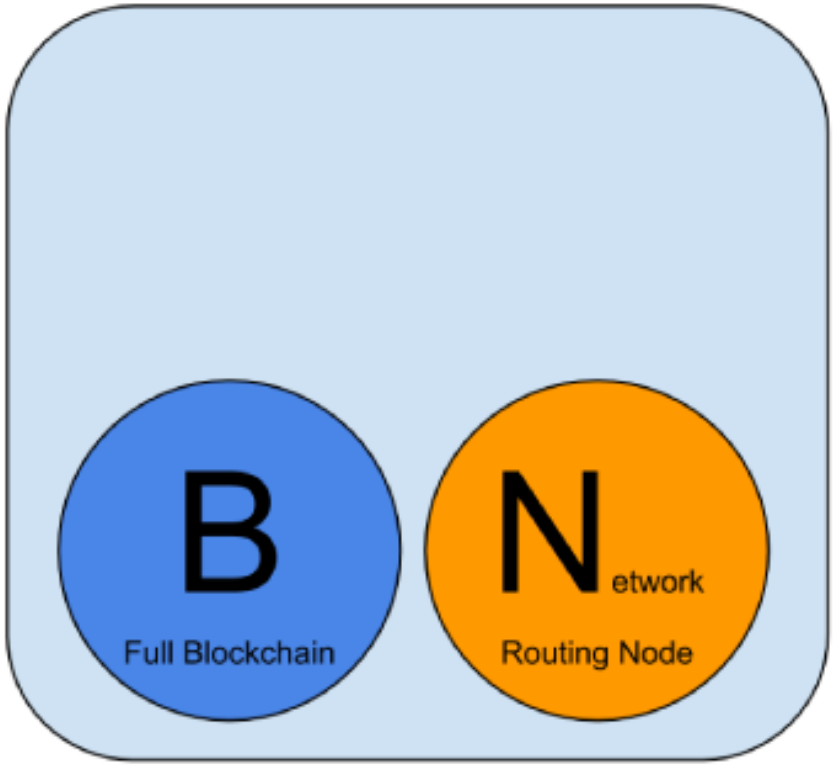


Reference Bitcoin Client

- A bitcoin network node
- with all four functions

Functions

- Wallet
- Miner
- Full blockchain database
- Network routing

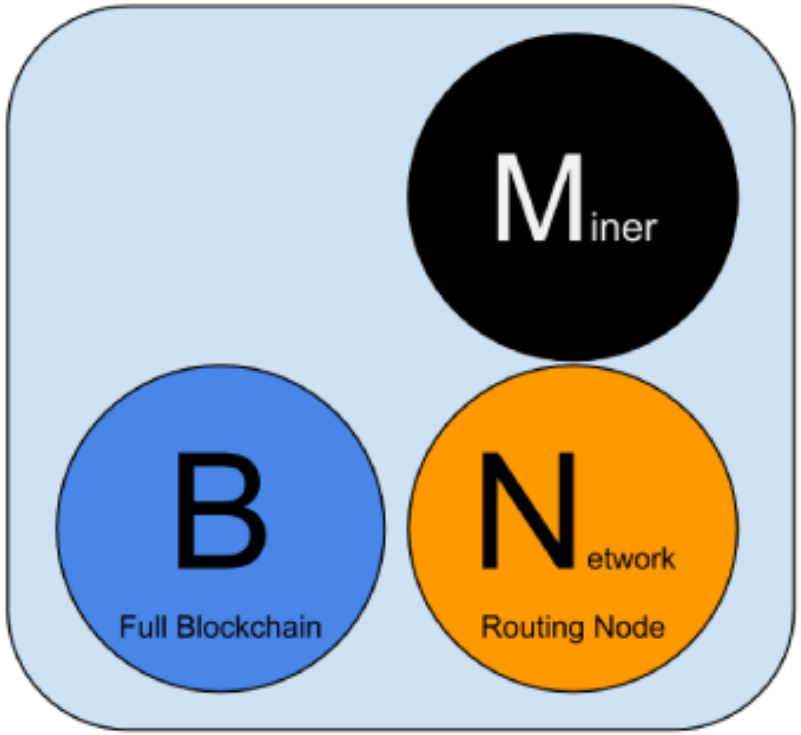


Full Block Chain Node

- A bitcoin network node
- with all two functions

Functions

- Full blockchain database
- Network routing

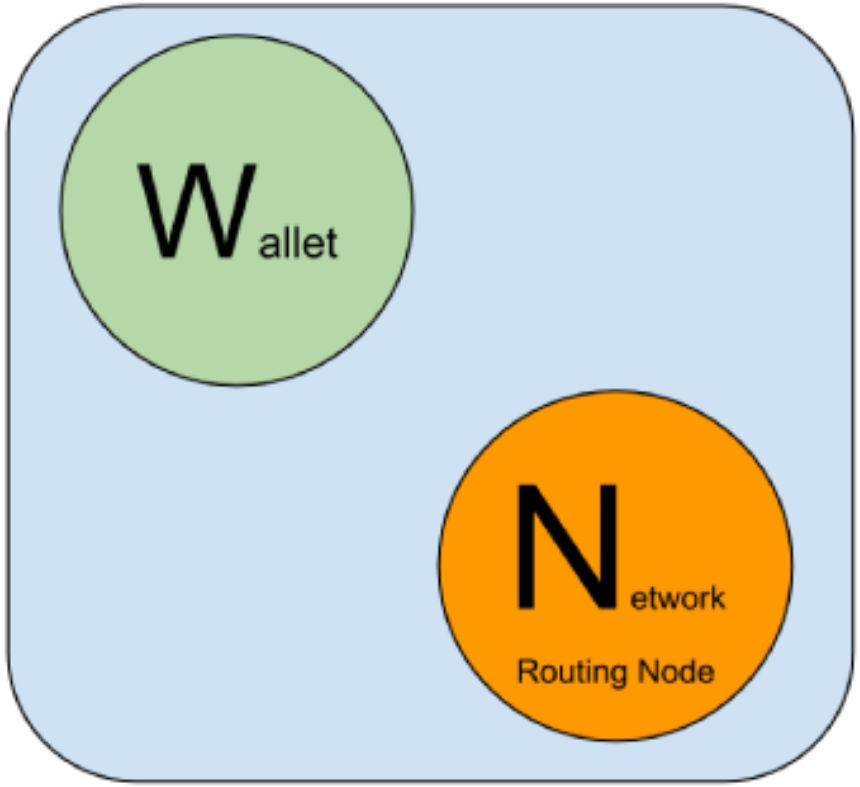


Solo Miner

- A bitcoin network node
- with three functions

Functions

- Miner
- Full blockchain database
- Network routing

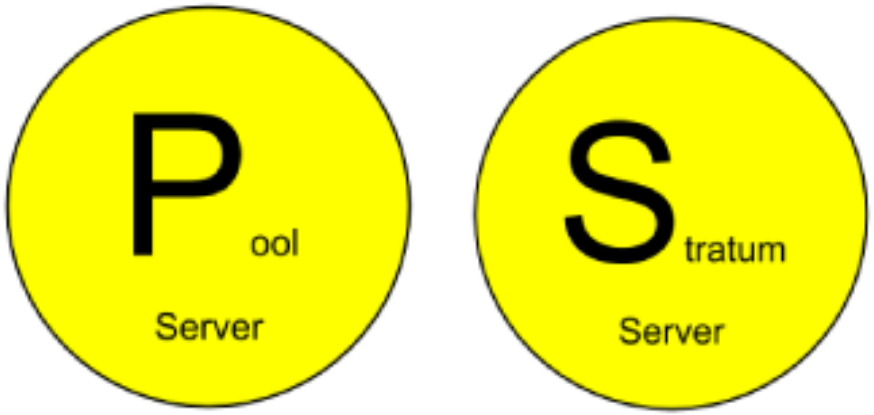


Lightweight Wallet

- A bitcoin network node
- with two functions

Functions

- Wallet
- Network routing

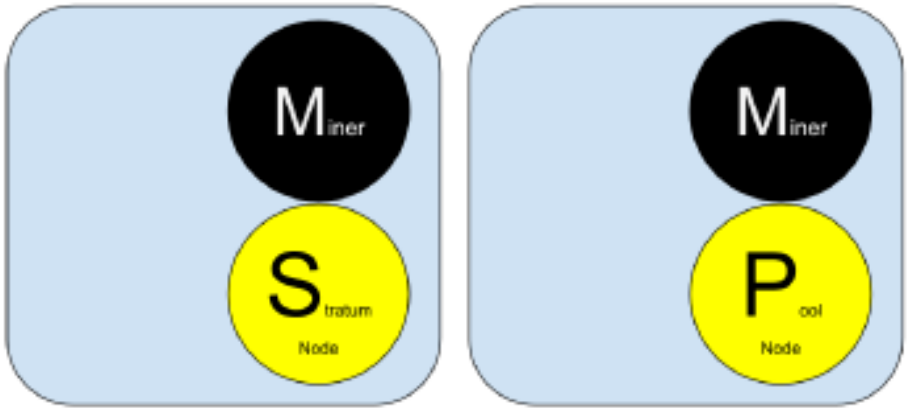


Pool Protocol Servers

- Gateway routers
- Connects Bitcoin to other protocols

Examples

- Stratum nodes
- Pool mining nodes



Mining Nodes

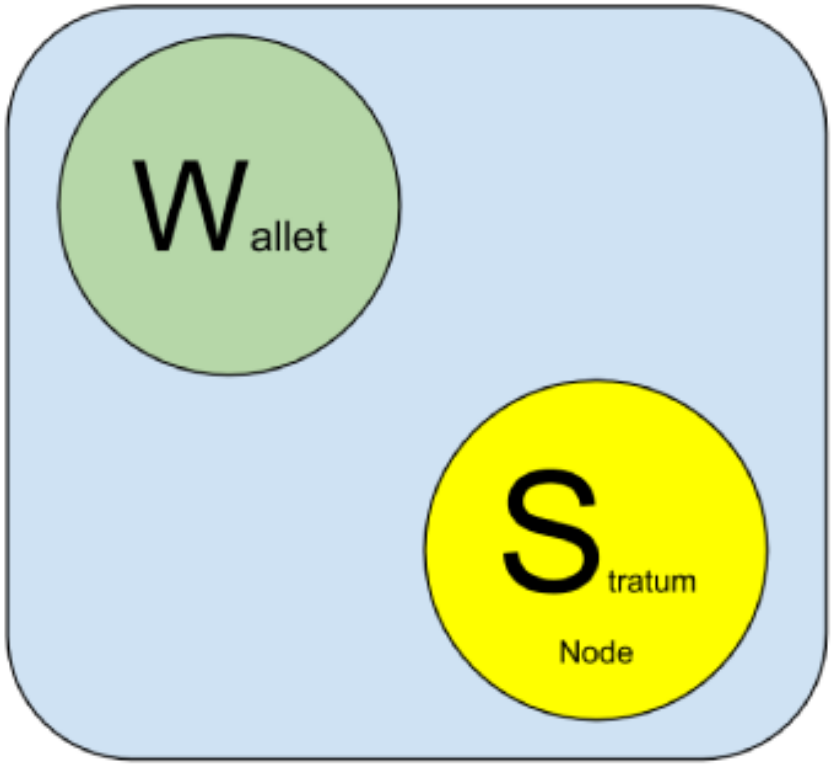
- Mining function
- without a blockchain

S

- Stratum protocol nodes

P

- Mining protocol node



Lightweight (SPV) Stratum Wallet

- Wallet

Functions

- Wallet
- Network node on the Stratum protocol



The Extended Bitcoin Network

Bitcoin Various Implementations



main bitcoin network

bitcoin P2P protocol

listening nodes



Bitcoin Core

5,000 to 8,000 nodes

various versions of **Bitcoin Core**



Other implementations

a few hundred nodes

various other implementations



Bitcoin P2P Protocol

All nodes run

without depending on the implementation

Other Versions

- Bitcoin Classic
- Bitcoin Unlimited
- BitcoinJ
- Libbitcoin
- btcd
- bcoin

Mining Nodes



https://news.bitcoin.com/block-propagation-startup-bloxroute-partners-with-mining-operation-rawpool/

mining nodes

- A small percentage of the nodes
- competing in the mining process
- validating transactions
- and creating new blocks

Large Companies Interface

Large companies

- running full-node clients
- based on the Bitcoin Core client

Content

- full copies of the blockchain
- network node
- No mining
- No wallet functions.

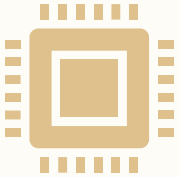
Edge routers

- These nodes act as
- network edge routers
- allowing various other services to be built on top

Services

- Exchanges
- Wallets
- block explorers
- merchant payment processing

Extended Network



The extended bitcoin network

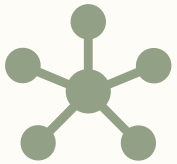
includes the network running the **bitcoin P2P protocol**



nodes

running **specialized protocols**

Attached to the main bitcoin P2P network are

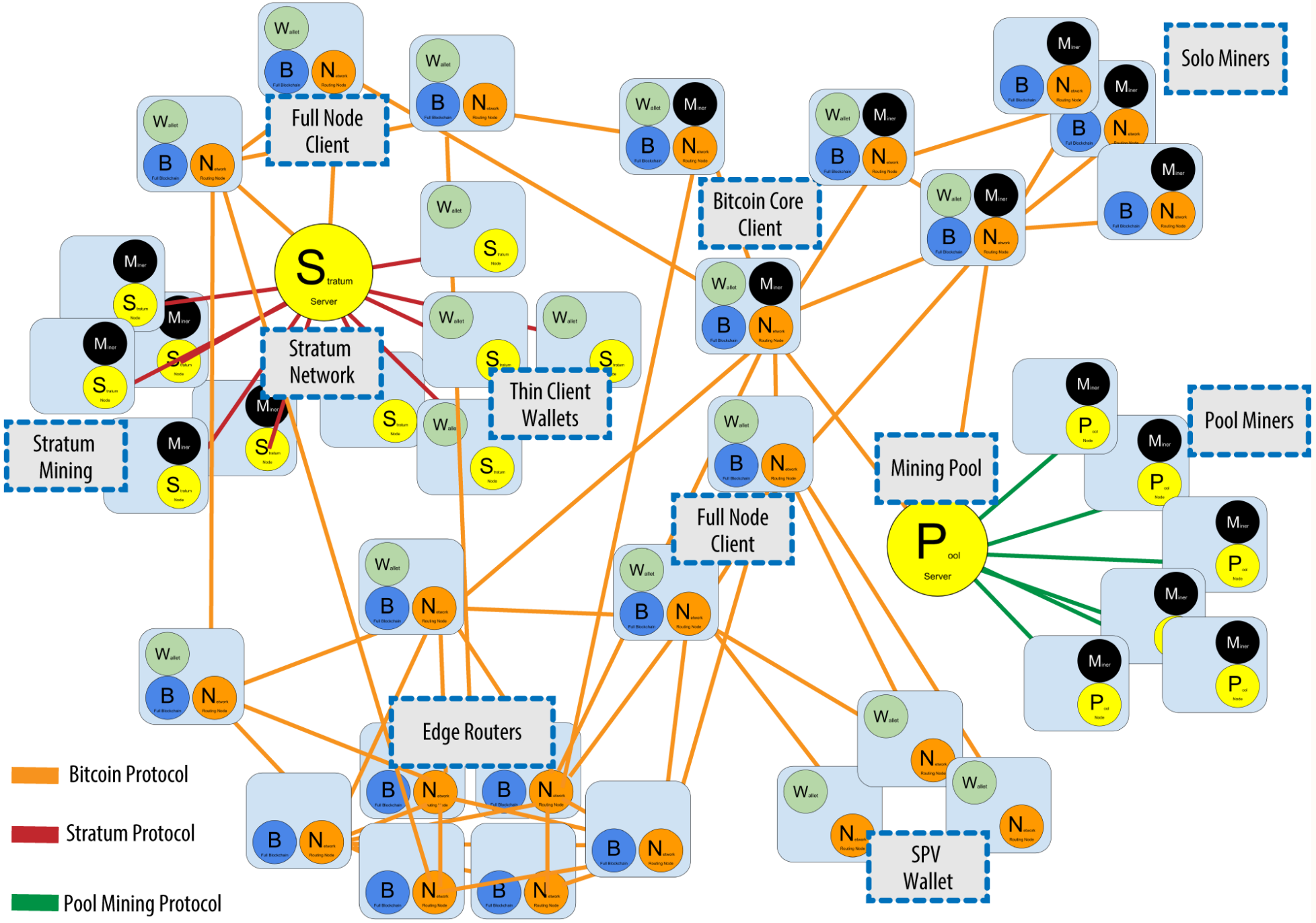


pool servers

protocol gateways

connect nodes **running other protocols**

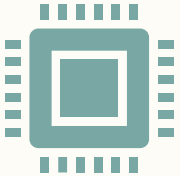
mostly pool mining nodes


Full Node Client
Solo Miners
Bitcoin Core Client
Stratum Server
Stratum Network
Thin Client Wallets
Stratum Mining
Mining Pool
Pool Server
Pool Miners
Full Node Client
Edge Routers
SPV Wallet
Bitcoin Protocol
Stratum Protocol
Pool Mining Protocol



Bitcoin Relay Networks

Bitcoin Relay Networks



High Latency

bitcoin P2P network
exhibits too high network latency



Bitcoin P2P for Mining

the specialized needs
of bitcoin mining nodes



Time-Sensitive Competition

Bitcoin miners are engaged
in a time-sensitive competition

Bitcoin Relay Networks



Minimize the Time

bitcoin miners must

minimize the time



Time frame

the propagation of a winning block

the beginning of the next round of competition



Latency vs Profit Margins

latency is directly related

to profit margins.

History



Mission: Minimize the Latency

A *Bitcoin Relay Network*

attempts to minimize the latency

in the transmission of blocks between miners.



2015

The original Bitcoin Relay Network

was created

in 2015

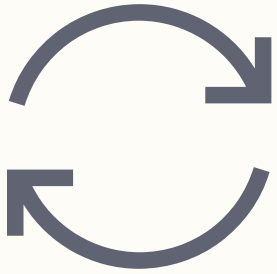


Matt Corallo 2015

core developer

developer of original Bitcoin Relay Network

Definition & History



Fast Synchronization

to enable fast synchronization

of blocks



low latency

between miners

with very low latency.

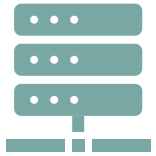
Definition & History



Host

on the Amazon Web Services infrastructure

around the world



Composition

The network consisted of

several specialized nodes



Connected

Majority of miners

and mining pools

Bitcoin Relay Networks: FIBRE



Matt Corallo, 2016

Creator of FIBRE

core developer

Replace

The original Bitcoin Relay Network

in 2016



FIBRE

Fast Internet Bitcoin Relay Engine

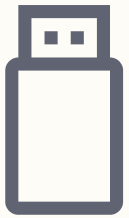
Bitcoin Relay Networks: FIBRE



UDP-based

relays blocks

within a network of nodes



Block Optimization

implements *compact block* optimization



reduce

data transmitted

reducing the network latency

Bitcoin Relay Networks: Falcon



Falcon

based on research at Cornell University

Another relay network

still in the proposal phase



cut-through-routing

Falcon uses “cut-through-routing”

instead of “store-and-forward”

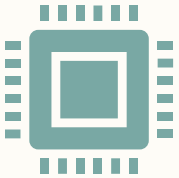


Propagate as Received

parts of blocks are propagated as they are received

rather than waiting until a complete block is received.

Relay Networks Role



No Replacements

Relay networks are not replacements

for bitcoin’s P2P network.



Additional connectivity

Instead they are overlay networks

provide additional connectivity

between nodes with specialized needs



Like freeways

that are not replacements for **rural roads**

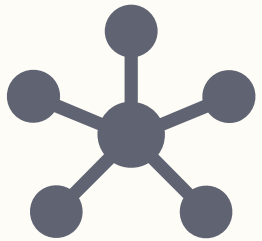
but rather shortcuts between two points with heavy traffic

you still need small roads to connect to the freeways.



Network Discovery

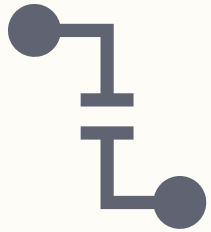
Network Discovery



Discover other Active Nodes

in order to participate a new node

discover other bitcoin nodes on the network



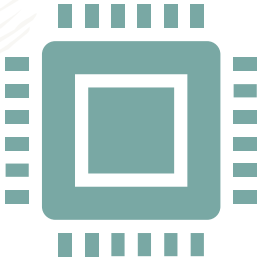
Boot up

a new node must

- discover
- connect

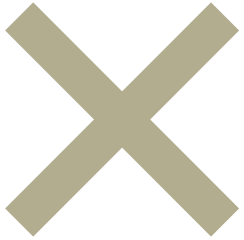
to at least one active node

Network Discovery



No Geographic Topology

the bitcoin network topology
is **not** geographically defined.



Random

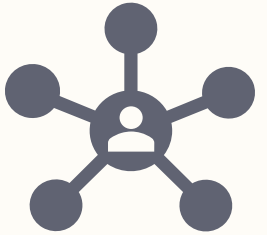
any existing bitcoin nodes
can be selected at random.

Connection to Known Peer



Port

To connect to a known peer
nodes establish a TCP connection
usually to **port 8333**
or an alternative port if one is provided



Handshake

Upon establishing a connection
the node will start a handshake
transmitting a version message

Version Message

nVersion	•The bitcoin P2P protocol version the client “speaks” •e.g., 70002
nLocalServices	•A list of local services supported by the node •currently just NODE_NETWORK
nTime	•The current time
addrYou	•The IP address of the remote node as seen from this node
addrMe	•The IP address of the local node, as discovered by the local node
subver	•A sub-version showing the type of software running on this node •e.g., /Satoshi:0.9.2.1/
BestHeight	•The block height of this node’s blockchain

Connection



First Message

The version message is always the first message

sent by any peer to another peer.



Check Compatibility

The local peer will examine the remote peer's reported nVersion

and decide if the remote peer is compatible.



Acknowledgement

If the remote peer is compatible,

the local peer will acknowledge the version message



Establish a connection

by sending a verack

How to Find Peers?



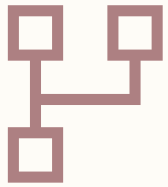
query DNS

The first method is to query DNS using a number of “DNS seeds,”



DNS seeds

Provide a list of IP addresses of bitcoin nodes.



Stable listening Nodes

Some of those DNS seeds provide a static list of IP addresses

of stable bitcoin listening nodes.

How to Find Peers?



BIND Nodes

Some of the DNS seeds are custom implementations of BIND



BIND

Berkeley Internet Name Daemon

return a random subset from a list of bitcoin node addresses

collected by a crawler or a long-running bitcoin node.

DNS Seeds

Bitcoin Core

The Bitcoin Core client contains

the names of 5 different DNS seeds

Reliability

the initial bootstrapping process depends on

the diversity of ownership

The diversity of implementation of the different DNS seeds

-dnsseed

the option to use the DNS seeds

set to 1 by default, to use the DNS seed

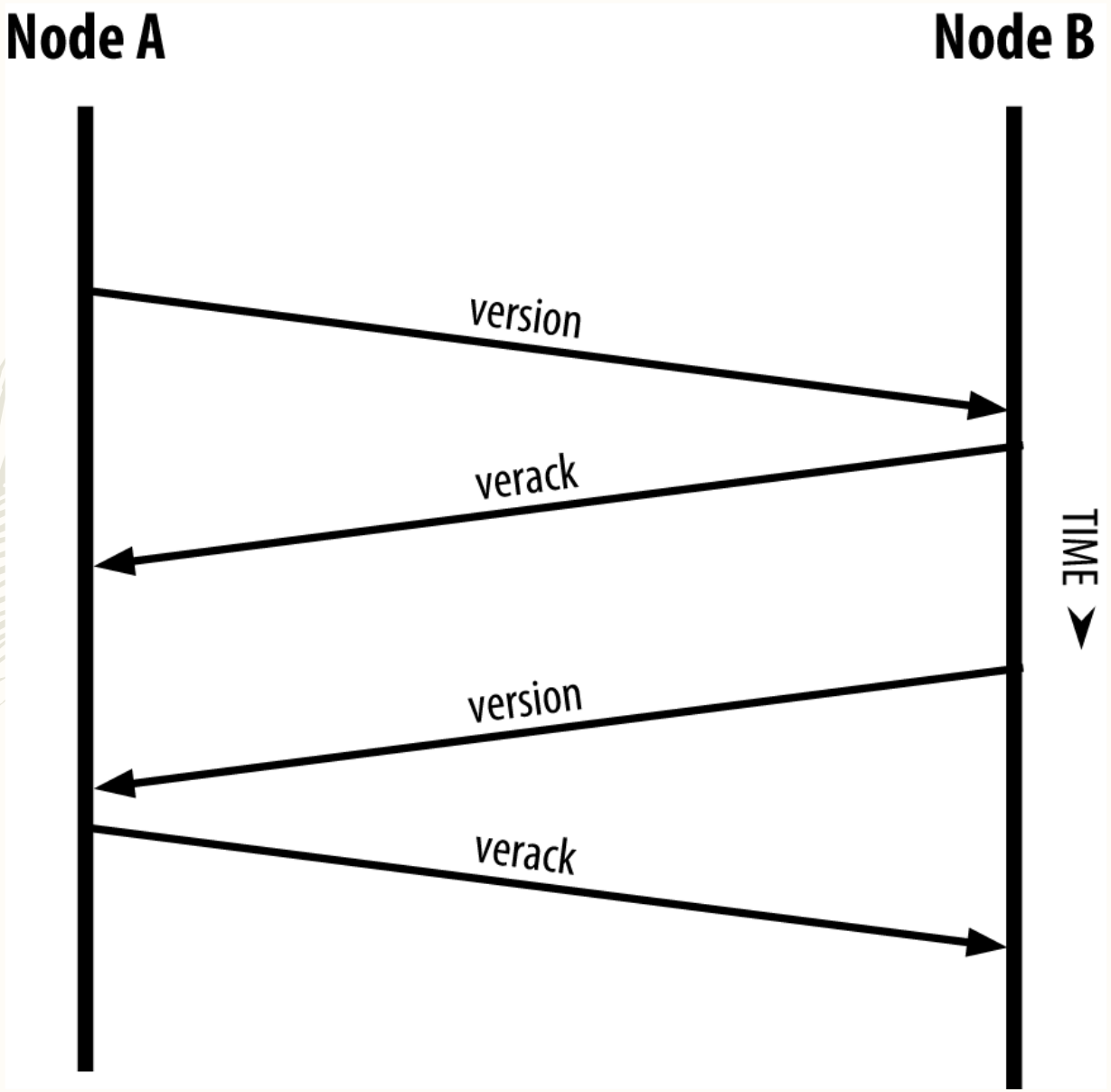


Figure 8-4. The initial handshake between peers

Alternative Approach



If nothing is known

a bootstrapping node that knows nothing of the network

must be given the IP address of at least one bitcoin node



-seednode

The command-line argument -seednode

can be used to connect to one node

just for introductions using it as a seed.



Using newl Peers

After the initial seed node is used to form introductions,

the client will disconnect from it

and use the newly discovered peers.

Forwarding the Address



Sending addr message

Once one or more connections are established

the new node will send an addr message

containing its own IP address to its neighbors.



Forward the message

The neighbors will, in turn,

forward the addr message to their neighbors,

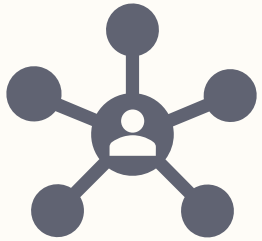


Better connections

ensuring that the newly connected node

becomes well known and better connected.

Advertisement of Existance



IP Addresses of other Peers

the newly connected node

can send getaddr to the neighbors,

asking them to return a list of IP addresses of other peers



Finding new Peers

That way, a node can find peers

to connect to and advertise its existence

on the network for other nodes to find it.

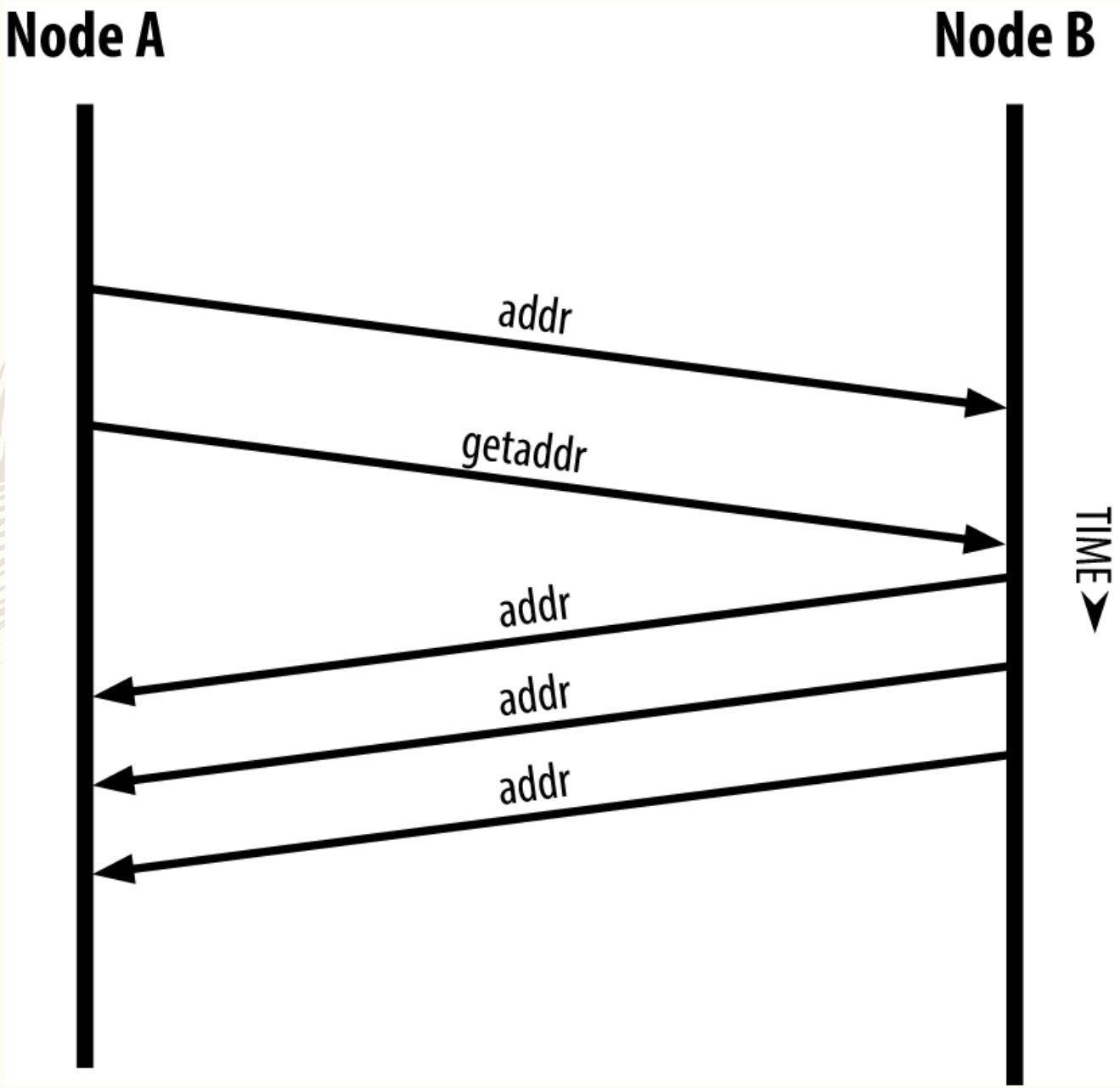


Figure 8-5. Address propagation and discovery

Organic Behaviour of Network



Maintain Connection

If there is no traffic on a connection,

nodes will periodically send a message to maintain the connection.



90 Minutes Off

If a node has not communicated for more than 90 minutes,

it is assumed to be disconnected

and a new peer will be sought.



Adjustments

Thus, the network dynamically adjusts

to transient nodes and network problems,



Organical Grow & Shrink

can organically grow and shrink

as needed without any central control.

What is a Bitcoin Node? - Step by Step Explanation



https://www.youtube.com/watch?v=sVeolsQ3cvU

Blockgeeks



Full Nodes

Full Nodes

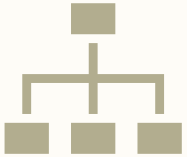


Full nodes

maintain a full blockchain with all transactions.

More accurately, they probably should be called

“full blockchain nodes.”



all full nodes

In the **early years** of bitcoin,

all nodes were full nodes



Bitcoin Core Client

Currently is a full blockchain node

Lightweight clients

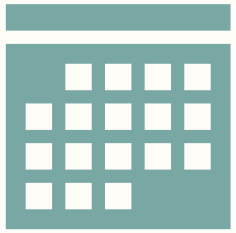


do not maintain a full blockchain



appeared in the past two years

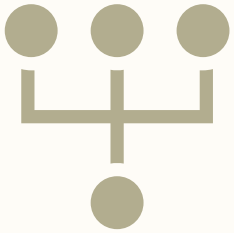
Advantages: Full Node



Complete blockchain

maintain a complete and up-to-date

copy of the bitcoin blockchain



Independent

they independently build and verify any transaction

without recourse or reliance on any other node or source of information

Disadvantages



Diskspace

to store the full blockchain

requires hundreds of gigabytes of persistent storage



Time to sync

it takes two to three days

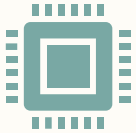
to sync to the network

Implementations

Alternatives	• There are a few alternative implementations • of full blockchain bitcoin clients
Programming languages	• Programmed in different programming languages • Built using different software architectures
The Satoshi client	• Identified as «Satoshi» in the sub-version string • Covers more than %75 of nodes on the network

Exchanging “Inventory”

Exchanging “Inventory”



Genesis block #0

statically embedded in the client software



New Full-node

brand-new node has no blockchain at all

it only knows one block, the genesis block



Start of sync

The process of syncing the blockchain

starts with the version message contains BestHeigth



BestHeight

a node’s current blockchain height

number of blocks

Catch Up



Node with longer blockchain

The peer that has the longer blockchain

has more blocks than the other node



Blocks for other nodes

The node can identify which blocks

the other node needs in order to “catch up.”



inv (inventory) message

identify the first 500 blocks

to share and transmit their hashes

Retrieving Blocks



Requesting blocks

by issuing a series of getdata messages

requesting the full block data



Hashes from inv message

identifying the requested blocks

using the hashes from the inv message.

Getting Off-line

This process	• sending getblocks, • gets an inv response, • downloading the missing blocks
Happens Any time	• this happens any time • a node goes offline
for any period of time	• for a few minutes and is missing a few blocks • or a month and is missing a few thousand blocks

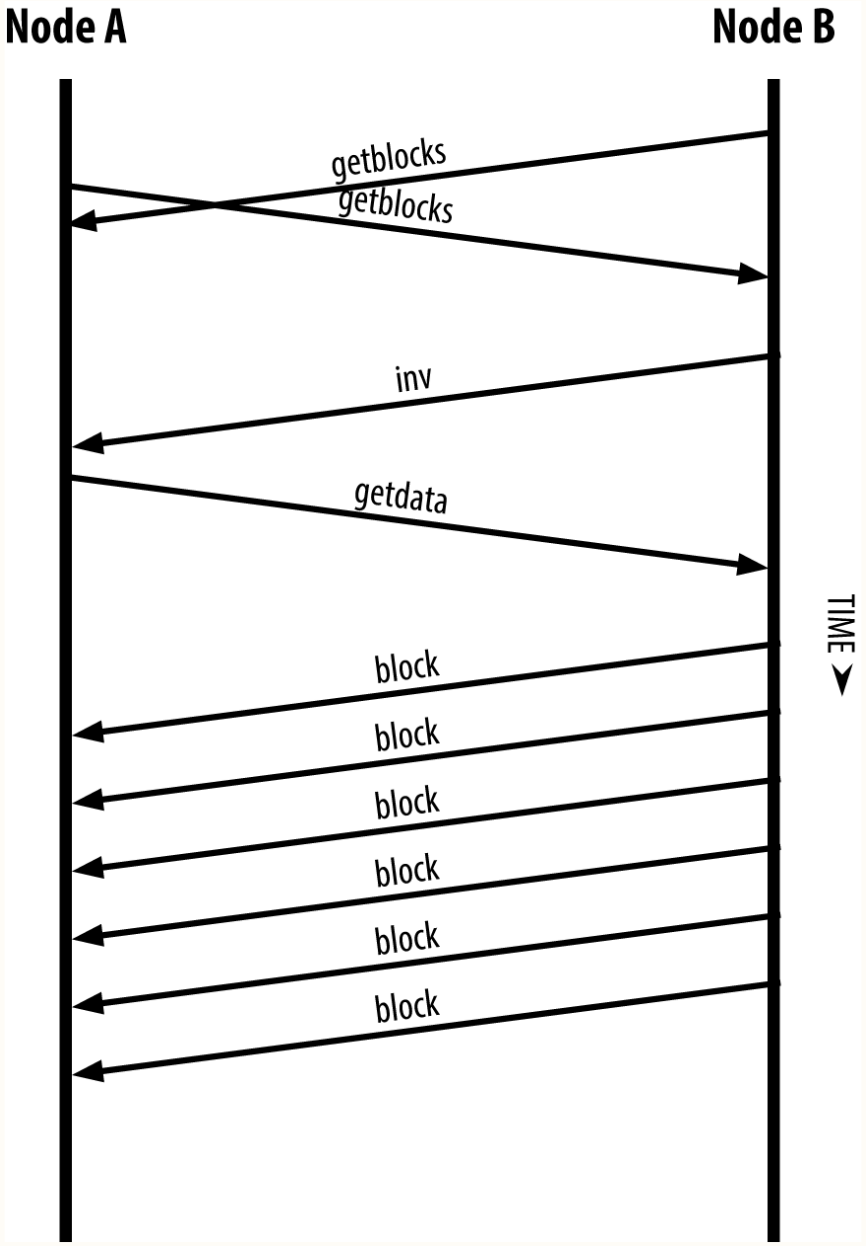


Figure 8-6. Node synchronizing the blockchain by retrieving blocks from a peer

Simplified Payment Verification (SPV) Nodes

SPV or Lightweight Clients



SPV

simplified payment verification



Constrained Devices

to run on space- and power-constrained devices

such as smartphones, tablets, or embedded systems



Becoming common

the SPV node is becoming the most common

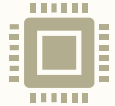
especially for **bitcoin wallets**



block headers

SPV nodes download only the block headers

and do not download the transactions included in each block.



smaller

without transactions chain is 1,000 times smaller

SPV



Validation of unspent

An SPV node cannot validate

whether the UTXO is unspent.



Merkle path

Instead, the SPV node will establish a link

between the transaction and the block that contains it,

using a *merkle path*



Block header

To get the block headers

SPV nodes use a getheaders message

instead of getblocks.



Respond

The responding peer will send

up to 2,000 block headers

using a single headers message.

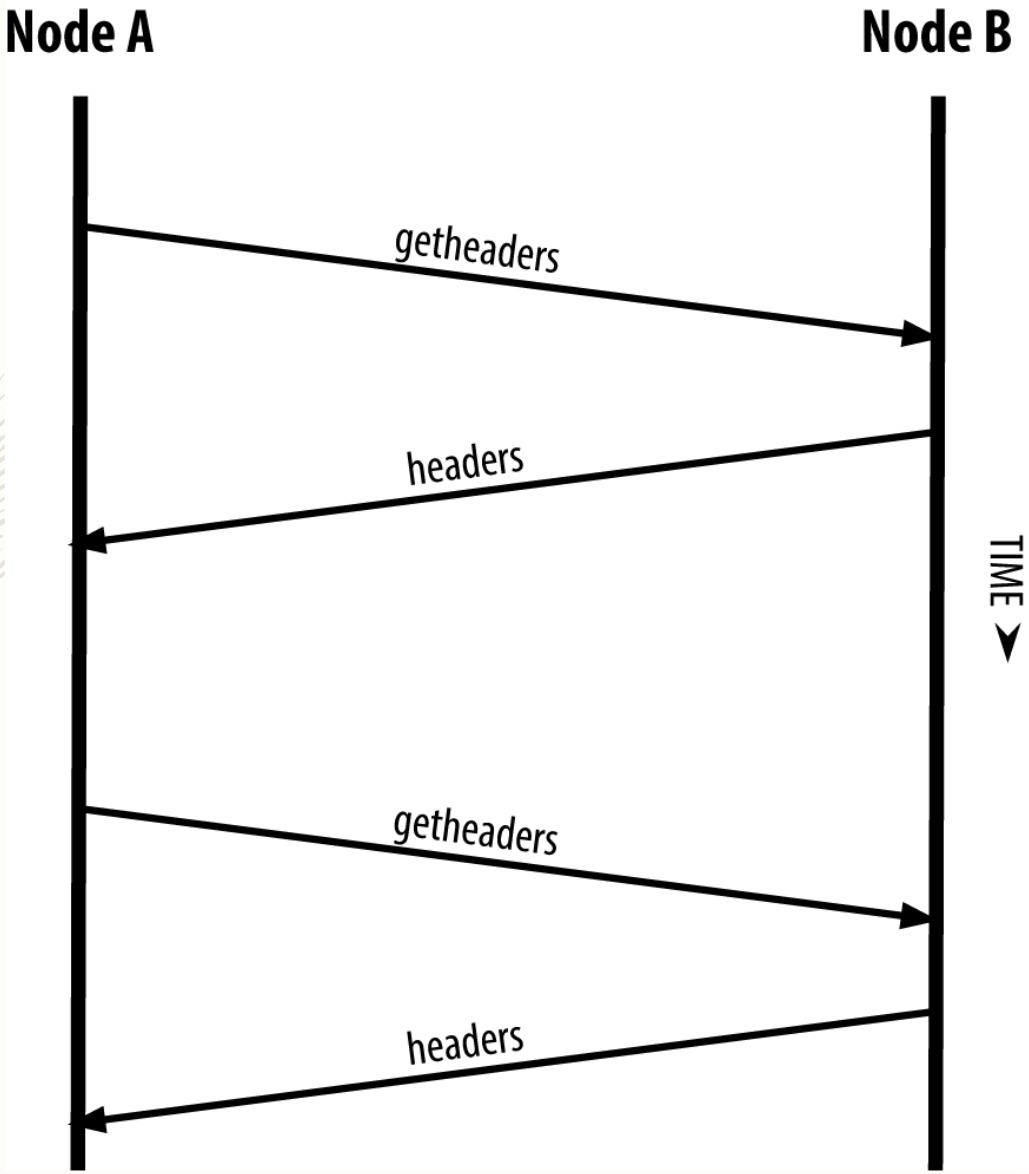


Figure 8-7. SPV node synchronizing the block headers



Bloom Filters

- How Bloom Filters Work
- How SPV Nodes Use Bloom Filters

Bloom Filters



Probabilistic filter

A bloom filter is a probabilistic search filter,

a way to describe a desired pattern

without specifying it exactly.



Efficient filters

Bloom filters offer an efficient way

to express a search pattern while protecting privacy.



Used by SPV Nodes

They are used by SPV nodes

to ask their peers

for transactions matching a specific pattern,



No reveal

without revealing exactly what they are searching for

which addresses, keys, or transactions.

Bloom Filters



Bloom filters serve allowing an SPV node

to specify a search pattern for transactions

that can be tuned toward precision or privacy.

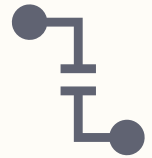


A more specific bloom filter

produce accurate results,

but revealing what patterns the SPV node is interested in,

thus revealing the addresses owned by the user's wallet



less specific bloom filter

produce more data with many irrelevant transactions

to the node, but will allow the node to maintain better privacy

What Are Bloom Filters?

https://www.youtube.com/watch?v=kfFacplFY4Y





Spanning Tree



SPV Nodes and Privacy

SPV Nodes and Privacy



SPV vs Full Nodes

Nodes that implement SPV

have weaker privacy than a full node



Full Nodes

A full node receives all transactions

and therefore reveals no information

about whether it is using some address in its wallet.



SPV Node reveals Info

An SPV node receives a filtered list of transactions

related to the addresses that are in its wallet.

Are Bloom Filters Sufficient?



Bloom filters

are a way to reduce the loss of privacy.



Without them

an SPV node would have to explicitly list the addresses it was interested in,

creating a serious breach of privacy.



Monitoring

an adversary monitoring the traffic of an SPV client

or connected to it directly as a node in the P2P network



Learn Addresses

can collect enough information over time

to learn the addresses in the wallet of the SPV client.



Encrypted and Authenticate Connections

- Tor Transport
- Peer-to-Peer Authentication and Encryption

Encrypted and Authenticated Connections



Clean Communication

Original implementations of bitcoin

Bitcoin communication was entirely clean



For Full Node

Not a problem



For SPV

A big problem



two solutions

Tor Transport

P2P Authentication and Encryption with BIP-150/151.

Tor Transport Layer



Tor

stands for *The Onion Routing network*



What is it?

is a software project and network



Offers

anonymity

Untraceability

privacy.

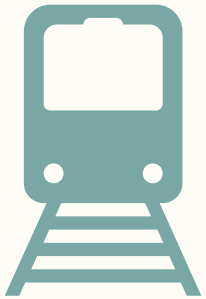


How?

encryption and encapsulation of data

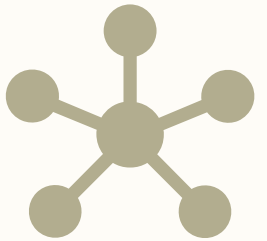
through randomized network paths

Bitcoin Options



Transport over Tor Network

Bitcoin Core offers configuration options
to transport traffic over the Tor network



Connect Tor Nodes

Bitcoin Core can also offer a Tor hidden service
allowing other Tor nodes
to connect to your node directly over Tor

Hidden Tor Service Offered Automatically

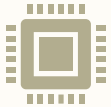


Works automatically if

Tor installed and

the Bitcoin Core process runs as a user with adequate permissions

to access the Tor authentication cookie.



Use the debug flag

to turn on Bitcoin Core's debugging

for the Tor service:



Command

$ bitcoind --daemon --debug=tor



You should see

"tor: ADD_ONION successful" in the logs,



indicating

Bitcoin Core has added a hidden service to the Tor network.

What is Tor and Should You Use It?



https://www.youtube.com/watch?v=6czcc1gZ7Ak

Mashable

Peer-to-Peer Authentication and Encryption



BIP-150 and BIP-151

Two Bitcoin Improvement Proposals



Benefits

add support for P2P authentication

encryption in the bitcoin P2P network.



BIPs define

optional services that may be offered

by compatible bitcoin nodes.

Peer-to-Peer Authentication and Encryption



BIP-151 enables

negotiated encryption for all communications

between two nodes that support BIP-151



BIP-150 offers

optional peer authentication



Authentication of identities

allows nodes to authenticate each other's identity

using ECDSA and private keys.



BIP-150 requires BIP-151

prior to authentication of the two nodes

encrypted communications as per BIP-151 should be established



Transaction Pools

Mempool



Unconfirmed transactions

Almost every node on the bitcoin network

maintains a temporary list of unconfirmed transactions

called the *memory pool*, *mempool*, or *transaction pool*.

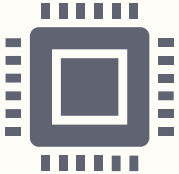


Maintain

Nodes use this pool

to keep track of transactions known

but are not yet included in the **blockchain**.



Incoming payments

For example, a wallet node

will use the transaction pool

to track incoming payments

Orphaned Transactions



Separate Pool

Some node implementations also maintain a separate pool of orphaned transactions.



Orphaned Transaction

If a transaction’s inputs refer to a transaction that is not yet known, such as a missing parent



Storage of orhpaned trancactions

the orphan transaction will be stored temporarily in the orphan pool until the parent transaction arrives.

Validation of Orphans



Check for orphans

When a transaction is added to the transaction pool

the orphan pool is checked for any orphans



Validation

Orphan is validated if any matching orphans

with reference to newly arrived transaction's outputs (its children)



Remove valid Orphans

If valid, orphans are removed from the orphan pool

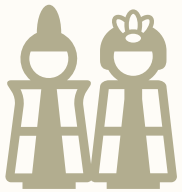
and added to the transaction pool

Validation of Orphans



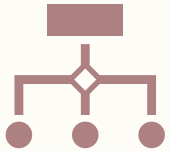
Complete the chain

complete the chain that started with the parent transaction.



Not Orphans anymore

the newly added transaction are not orphans any more



Recursive processing

the process is repeated recursively looking for any further descendants,

until no more descendants are found.

UTXO Database



Implementation with UTXO

Some implementations of the bitcoin client

also maintain a UTXO database or pool



Unspent outputs

the set of all unspent outputs on the blockchain.



Millions of Entries

Contains millions of entries of unspent transaction outputs



All the way back to genesis block

everything that is unspent

from all the way back to the genesis block.

Discussions



Your opinion

What do you think on this chapter?

Challenging

What is the most interesting point?

Clarification

What are the unclarified points?

Critical issue

Select and explain a critical issue in this chapter